BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Debugging and Nested Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does sequence mean in programming?

2. What is 4 + 3 * 2?
   A) 10
   B) 14
   C) 20
   D) 24

3. What is 2 ^ 3 in OCR ERL?

4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

Today's objectives

Identify and fix syntax and logic errors. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Syntax error

An incorrect language keyword or missing ENDIF breaks the grammar.

Logic error

A running program may produce an incorrect result because its steps are wrong.

Nested selection

Each IF within an IF needs the correct closing ENDIF.

Debugging cycle

Test, locate, correct and retest the fault.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Syntax error
An incorrect language keyword or missing ENDIF breaks the grammar. For example, consider a membership checker.

Logic error
A running program may produce an incorrect result because its steps are wrong. For example, consider a sports bands.

Nested selection
Each IF within an IF needs the correct closing ENDIF. For example, consider a ticket discounts.

Debugging cycle
Test, locate, correct and retest the fault. For example, consider a club rules.

I Do - Worked model

A membership checker is being developed. Explain how syntax error and logic error are used. [4 marks]

- Identify where this particular program needs control or values.
- Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].
- Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A sports bands is being programmed. Explain why syntax error matters. [2 marks]

2. For a ticket discounts, explain how logic error is used. [2 marks]

3. A club rules needs nested selection and debugging cycle. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the logical operator so 260 is rejected.

```
octet = 260
if octet >= 0 OR octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how logic error and nested selection can help a ride height. [4 marks]

Q4. Explain the roles of syntax error, nested selection and debugging cycle when creating a reward scoring. [6 marks]

Exit Ticket - True or False?

An incorrect language keyword or missing ENDIF breaks the grammar.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Instructions execute in the specified order.

2. A - 10

3. 8; ^ denotes exponentiation in OCR ERL.

4. Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly.

I Do - Explain the model

Identify where this particular program needs control or values.

Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

3. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

Code First - Answer

```
octet = 260
if octet >= 0 AND octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

Expected output: Invalid

You Do - Self-mark

Q2: Syntax error [1]; Nested selection [1].

Q3: Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1].

Q4: Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.